

Summary

I am a senior undergraduate in Engineering Mechanics at Zhejiang University, working under the guidance of Prof. Shiyong Xiong. My research lies at the intersection of scientific machine learning and scientific computing, focusing on neural operators and computational methods for PDEs, with applications to fluid systems, Schrödinger-type models, and reduced-order modeling.

Education

- Zhejiang University, Engineering Mechanics** Sep 2023 – Present
GPA: 90.2% | Ranking: 1/55

Publications and Preprints

- Local gradient neural operator** *Preprint*
Baiming Zhang, Jinsong Tang, Ying Xu*, Lihua Chen, Shiyong Xiong*
- Learnable composition for neural operators** *Preprint*
*Zituo Chen, Baiming Zhang, Sili Deng**
- Gauge-Canonical Eigenfluids: Geometry-Conditioned Neural Spectra** *Preprint*
Baiming Zhang, Zhecheng Wang, Shiyong Xiong, Eitan Grinspun*

Research Experience

- Machine Learning in Fluid and PDE Systems** Jan 2025 – Present
Advisor: Prof. Shiyong Xiong, Zhejiang University.
Scientific machine learning, neural operators, PDE learning, reduced-order fluid modeling, and physics-constrained inverse problems.
- Quantum States & Devices Research** Oct 2023 – Dec 2024
Under Prof. Dawei Wang and Prof. Xingqi Xu.
Rydberg atoms, Raman effects, EIT, Fock states, superlattices, and related theoretical modeling.

Selected Awards and Honors

- International College Student Engineering Mechanics Competition, World Gold Medal**
First student in ZJU-SAA history to win the gold medal.
- Chinese National Zhou Peiyuan Mechanics Competition, First Prize
- National Mechanics “101 Plan” Undergraduate Academic Forum, Outstanding Oral Presentation
- Mathematics Competition for College Students, First Prize
- Government Scholarship

Other Experience

- Teaching Assistant, Chinese Physics Olympiad Training Camp, 2024
- Awarded “Five-Star Volunteer” — the university’s highest honor for public service
- Volunteer Coordinator, 2024 International Symposium on Quantum Computing and Quantum Optics

Technical Skills

- Programming & Scientific Computing:** Python, MATLAB, PyTorch, LaTeX
- Tools:** GitHub, Houdini, SolidWorks, Raspberry Pi
- Machine Tools:** Lathe, milling machine, shaper, slotter